

OutLine

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Note : This file covers these topics.

CHAPTER 1: INTRODUCTION

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- Performance: Delay, Loss and Throughput in Computer Networks
- Protocol layers and Their Service Models
- Networks under attack
- History of Computer Networking

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- Whats Inside a Router?
- The Internet Protocol
- Generalized Forwarding
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Chapter 1: Introduction — MCQs

1. What is the Internet from a “nuts-and-bolts” view?

- A. A single computer network
- B. A network of networks
- C. A social media platform only
- D. A search engine

Answer: B

2. Which of the following is part of the Internet’s “services” view?

- A. A platform for building network applications
- B. A packet switch
- C. A fiber optic cable
- D. A router table

Answer: A

3. Which human activity is an example of a protocol?

- A. Sleeping
- B. Asking someone for the time and receiving an answer
- C. Walking alone
- D. Reading silently

Answer: B

4. Which access technology is wired and can support up to hundreds of Gbps per link?

- A. 4G LTE
- B. WiFi
- C. Ethernet
- D. DSL

Answer: C

5. Which access technology is wireless and supports tens to hundreds of Mbps per device?

- A. Ethernet
- B. 802.11 WiFi
- C. DSL
- D. Cable access network

Answer: B

6. Which physical layer technology has the highest transmission rate and lowest bit error rate in practice?

- A. Coaxial cable
- B. Fiber optic cable
- C. DSL
- D. 4G LTE

Answer: B

7. What is forwarding?

- A. Determining the full path from source to destination
- B. Moving packets from a router's input link to the correct output link
- C. Encrypting packets before sending
- D. Creating network applications

Answer: B

8. What is routing?

- A. Moving packets locally inside a router
- B. Determining source-to-destination paths taken by packets
- C. Checking packet errors only
- D. Transmitting bits into a link

Answer: B

9. Which switching technique is used in the Internet?

- A. Circuit switching
- B. Packet switching
- C. Telephone switching only
- D. Manual switching

Answer: B

10. Which of the following is a characteristic of packet switching?

- A. Resources are reserved before communication begins
- B. Resources are used on demand
- C. It has no congestion
- D. It guarantees fixed delay

Answer: B

11. Which of the following is a characteristic of circuit switching?

- A. Resources are reserved for a call
- B. Packets are always queued
- C. It is used only in WiFi
- D. It has variable end-to-end delay

Answer: A

12. Which multiplexing techniques are used in circuit switching?

- A. HTTP and FTP
- B. FDM and TDM
- C. TCP and UDP
- D. DSL and LTE

Answer: B

13. If there are 20 circuits between A-B, 19 between B-C, 15 between C-D, and 16 between D-A, what is the maximum number of ongoing connections?

- A. 35
- B. 50
- C. 70
- D. 100

Answer: C

14. What does it mean that the Internet is a “network of networks”?

- A. It is made of only one global network
- B. It is made of many interconnected networks
- C. It is only a collection of websites
- D. It has no access networks

Answer: B

15. In a 10 Mbps channel with 5 users, if each user generates traffic at an average rate of 2 Mbps and transmits at 2 Mbps, which switching method works best?

- A. Packet switching
- B. Circuit switching
- C. Neither
- D. Both always fail

Answer: B

16. In a 10 Mbps channel with 5 users, if each user averages 0.21 Mbps but transmits at 15 Mbps when active, which method is better?

- A. Circuit switching
- B. Packet switching
- C. Neither
- D. Frequency switching

Answer: B

17. What is processing delay?

- A. Time spent waiting in packet buffers
- B. Time needed to check and process packet information in a router
- C. Time needed for bits to propagate through a link
- D. Time needed to transmit packet bits into the link

Answer: B

18. What is queueing delay?

- A. Time spent waiting in packet buffers
- B. Time spent checking packet headers
- C. Time spent propagating through the medium
- D. Time spent creating an application message

Answer: A

19. What is transmission delay?

- A. Time needed for bits to physically travel through the medium
- B. Time spent waiting in a queue
- C. Time spent transmitting packet bits into the link
- D. Time spent routing packets globally

Answer: C

20. What is propagation delay?

- A. Time needed for bits to physically travel through the transmission medium
- B. Time needed to inspect packet contents
- C. Time spent in router buffers
- D. Time needed to add headers

Answer: A

21. A packet is 1500 bytes and the link rate is 1 Gbps. What is the transmission delay?

- A. 0.000012 seconds
- B. 0.000096 seconds
- C. 0.0033 seconds
- D. 8×10^{-5} seconds

Answer: A

22. A packet is 1200 bytes and the link rate is 100 Mbps. What is the transmission delay?

- A. 0.000012 seconds
- B. 0.000096 seconds
- C. 0.0033 seconds
- D. 1 second

Answer: B

23. If a packet length is 8000 bits, what is the transmission delay at link 2?

- A. 8×10^{-5} seconds
- B. 0.0033 seconds
- C. 1.5 Mbps
- D. 50 Mbps

Answer: A

24. What is the propagation delay along link 2?

- A. 0.000012 seconds
- B. 8×10^{-5} seconds
- C. 0.0033 seconds
- D. 0.56 seconds

Answer: C

25. What is the maximum throughput in the given sender-receiver scenario?

- A. 1.5 Mbps
- B. 50 Mbps
- C. 90 Mbps
- D. 300 Mbps

Answer: A

26. In the four-server/four-client scenario, what is the maximum achievable end-to-end throughput for each pair?

- A. 30 Mbps
- B. 50 Mbps
- C. 75 Mbps
- D. 90 Mbps

Answer: B

27. What is the utilization of the server links with transmission capacity RS?

- A. 0.50
- B. 0.56
- C. 0.67
- D. 1.00

Answer: D

28. What is the utilization of the shared link with capacity R?

- A. 0.50
- B. 0.56
- C. 0.67
- D. 1.00

Answer: C

29. What is the utilization of the client links with transmission capacity RC?

- A. 0.56
- B. 0.67
- C. 0.90
- D. 1.00

Answer: A

30. Which layer contains protocols that are part of distributed network applications?

- A. Physical layer
- B. Link layer
- C. Transport layer
- D. Application layer

Answer: D

31. Which layer transfers data between processes on different hosts?

- A. Application layer
- B. Transport layer
- C. Network layer
- D. Physical layer

Answer: B

32. Which layer delivers datagrams from a source host to a destination host?

- A. Link layer
- B. Network layer
- C. Application layer
- D. Physical layer

Answer: B

33. Which layer transfers data between neighboring network devices?

- A. Transport layer
- B. Network layer
- C. Link layer
- D. Application layer

Answer: C

34. Which layer transfers bits into and out of the transmission medium?

- A. Physical layer
- B. Link layer
- C. Network layer
- D. Transport layer

Answer: A

35. What is the unit of data at the application layer?

- A. Segment
- B. Datagram
- C. Message
- D. Frame

Answer: C

36. What is the unit of data at the transport layer?

- A. Bit
- B. Frame

- C. Segment
- D. Datagram

Answer: C

37. What is the unit of data at the network layer?

- A. Datagram
- B. Message
- C. Segment
- D. Bit

Answer: A

38. What is encapsulation?

- A. Removing all headers from a packet
- B. Taking data from the layer above and adding this layer's header
- C. Encrypting all packets
- D. Sending data without protocols

Answer: B

39. Which security defense provides confidentiality by encoding contents?

- A. Firewall
- B. Encryption
- C. Access control
- D. Authentication

Answer: B

40. Which security defense filters or blocks traffic using middleboxes?

- A. Firewall
- B. Encryption
- C. Digital signature
- D. Authentication

Answer: A

41. Which security defense is used to detect message tampering and identify the sender?

- A. Firewall
- B. Digital signature

- C. Access control
- D. Packet switching

Answer: B

42. Which security defense limits use of resources to specific users?

- A. Authentication
- B. Access control
- C. Encryption
- D. Propagation

Answer: B

43. Which security defense proves that you are who you say you are?

- A. Authentication
- B. Firewall
- C. Forwarding
- D. Encapsulation

Answer: A

44. When did early studies of packet switching by Baran, Davies, and Kleinrock occur?

- A. Early 1960s
- B. Late 1960s
- C. Early 1980s
- D. 1990s

Answer: A

45. When was the first ARPAnet node operational?

- A. Early 1960s
- B. Late 1960s
- C. 1970s
- D. 2000–2010

Answer: B

46. When was IP standardized in RFC 791?

- A. Early 1960s
- B. 1970s
- C. Early 1980s
- D. Late 1980s

Answer: C

47. When was congestion control added to TCP?

- A. Early 1980s
- B. Late 1980s
- C. 1990s
- D. 2010–2020

Answer: B

48. When did the World Wide Web start up?

- A. 1970s
- B. Early 1980s
- C. 1990s
- D. 2000–2010

Answer: C

Chapter 2: Application Layer — MCQs

1. Which characteristic is associated with the client-server architecture?

- A. No server is always on
- B. A server is always on
- C. All peers have equal roles only
- D. No fixed IP address is used

Answer: B

2. Which application protocol commonly uses the client-server model?

- A. HTTP
- B. DNS only
- C. Ethernet
- D. IP

Answer: A

3. Which characteristic is associated with P2P architecture?

- A. There is always an always-on server
- B. Processes only request services
- C. A process can request and provide service
- D. It requires a well-known server IP address

Answer: C

4. What transport service does UDP provide?

- A. Reliable delivery
- B. Flow control
- C. Best-effort delivery
- D. Guaranteed ordering

Answer: C

5. Which service is provided by TCP?

- A. Loss-free data transfer
- B. Best-effort only
- C. No congestion control
- D. No flow control

Answer: A

6. What does TCP flow control prevent?

- A. Router overflow only
- B. Receiver buffer overflow
- C. IP address conflict
- D. DNS caching

Answer: B

7. What does it mean that HTTP is stateless?

- A. HTTP has no status codes
- B. The server does not remember previous interactions
- C. HTTP does not use TCP
- D. HTTP cannot use cookies

Answer: B

8. What is an HTTP cookie used for?

- A. To encrypt a packet
- B. To allow a server to access stored information about earlier browser interactions
- C. To replace DNS
- D. To close a TCP connection

Answer: B

9. What is the purpose of an HTTP GET request?

- A. To send email
- B. To request an object from a web server
- C. To assign an IP address
- D. To encrypt web content

Answer: B

10. What is the purpose of a conditional HTTP GET?

- A. To request an object only if it has changed
- B. To always force a new download
- C. To close the connection
- D. To delete a cached object

Answer: A

11. In the request line GET /... HTTP/1.1, what version of HTTP is being used?

- A. HTTP/1.0
- B. HTTP/1.1
- C. HTTP/2
- D. HTTP/3

Answer: B

12. Which HTTP header indicates that the client has a cached copy?

- A. Host
- B. Accept
- C. If-Modified-Since
- D. User-Agent

Answer: C

13. If an HTTP response uses HTTP/1.0 and has Connection: Close, what will the server do?

- A. Keep the TCP connection open forever
- B. Close the TCP connection after sending the response
- C. Switch to UDP
- D. Send a DNS request

Answer: B

14. What is an advantage of a web cache?

- A. It increases bandwidth use
- B. It can reduce page load time
- C. It disables HTTP
- D. It prevents TCP connections

Answer: B

15. Which feature was introduced by HTTP/2?

- A. No use of TCP
- B. Interleaving pieces of large and small objects
- C. Removal of response codes
- D. Mandatory cookies

Answer: B

16. Which information appears in an HTTP response message?

- A. Response code
- B. MAC address only

- C. DHCP offer
- D. DNS root server address

Answer: A

17. How many RTTs occur before a client can begin sending the email message after TCP and SMTP handshaking?

- A. 1
- B. 2
- C. 3
- D. 4

Answer: C

18. Which characteristic applies to HTTP only?

- A. Client pull protocol
- B. Uses server port 25
- C. Pulls email from server
- D. Uses only UDP

Answer: A

19. Which characteristic applies to SMTP only?

- A. Uses server port 80
- B. Operates mostly as a client push protocol
- C. Used for web browsing
- D. Uses HTTP GET

Answer: B

20. Which characteristic applies to both HTTP and SMTP?

- A. ASCII command/response interaction
- B. Uses UDP only
- C. Provides IP addresses
- D. Uses port 25 only

Answer: A

21. Which protocol pushes email from a mail client to a mail server?

- A. IMAP
- B. SMTP
- C. DNS
- D. HTTP

Answer: B

22. Which protocol pulls email from a mail server to a mail client?

- A. IMAP
- B. SMTP
- C. IP
- D. UDP

Answer: A

23. What does an authoritative DNS server provide?

- A. Definitive hostname-to-IP mappings
- B. TCP congestion control
- C. HTTP caching
- D. Packet forwarding

Answer: A

24. What does a local DNS server do?

- A. Only stores web pages
- B. Replies to DNS queries by contacting other DNS servers
- C. Sends email messages
- D. Creates TCP sockets

Answer: B

25. What is the role of a TLD DNS server?

- A. Responsible for domains such as .com or .edu
- B. Encrypts HTTP requests
- C. Assigns MAC addresses
- D. Controls TCP retransmission

Answer: A

26. What is the value of DNS caching?

- A. Slower replies
- B. Faster replies and less DNS load
- C. More packet loss
- D. Larger TCP headers

Answer: B

27. What does a DNS type A record contain?

- A. Hostname and IP address
- B. Email body
- C. TCP sequence number
- D. HTTP cookie value only

Answer: A

28. Which requests required 8 time units in the DNS caching scenario?

- A. $t = 0$, $t = 1$, and $t = 30$
- B. $t = 5$ and $t = 12$
- C. $t = 10$ only
- D. All requests

Answer: A

29. Which request required 2 time units in the DNS caching scenario?

- A. $t = 0$
- B. $t = 1$
- C. $t = 10$
- D. $t = 30$

Answer: C

30. Which caching types benefit from other users' recent requests?

- A. Local DNS caching and HTTP local web caching
- B. Browser caching only
- C. TCP caching only
- D. UDP caching only

Answer: A

31. What is a manifest file used for in video streaming?

- A. To assign an IP address
- B. To tell the client where to retrieve video segments at different rates
- C. To calculate checksums
- D. To terminate TCP connections

Answer: B

32. How does a CDN serve many simultaneous users?

- A. By storing content at one central server only
- B. By storing multiple copies at geographically distributed sites
- C. By disabling caching
- D. By using only SMTP

Answer: B

33. In DASH, video chunks are stored how?

- A. Only at one fixed quality
- B. Encoded at multiple rates
- C. Without a manifest file
- D. Only as email attachments

Answer: B

34. What does DASH allow the client to do?

- A. Adapt video quality to available bandwidth
- B. Replace TCP with DNS
- C. Prevent all packet loss
- D. Disable HTTP

Answer: A

35. Which socket type is created by `socket(AF_INET, SOCK_DGRAM)`?

- A. TCP socket
- B. UDP socket
- C. HTTP socket
- D. DNS socket

Answer: B

36. Which socket type is created by `socket(AF_INET, SOCK_STREAM)`?

- A. TCP socket
- B. UDP socket
- C. IP socket only
- D. Ethernet socket

Answer: A

37. What does a UDP application need to specify when sending data?

- A. Only the object name
- B. Destination IP address and port number
- C. Only the DNS name
- D. Only the MAC address

Answer: B

38. What does TCP provide to applications?

- A. Reliable, in-order byte-stream transfer
- B. Unreliable datagram service
- C. No connection setup
- D. No byte ordering

Answer: A

39. If a web server has five ongoing TCP connections on port 80, how many TCP sockets are in use at the server?

- A. 1
- B. 5
- C. 6
- D. 80

Answer: C

40. What happens when `connect()` is called?

- A. A server deletes a socket
- B. A new client socket is created and connected to a specified server
- C. DNS cache is cleared
- D. UDP becomes reliable

Answer: B

Chapter 3: Transport Layer — MCQs

1. Where is transport-layer functionality primarily implemented?

- A. In routers only
- B. At hosts at the network edge
- C. In DNS servers only
- D. In switches only

Answer: B

2. The transport layer provides host-to-host delivery service.

- A. True
- B. False

Answer: A

3. Which service is provided by TCP?

- A. In-order data delivery
- B. Message boundary preservation
- C. Unreliable datagram transfer
- D. No congestion control

Answer: A

4. Which service is provided by UDP?

- A. Reliable byte stream
- B. Message abstraction preserving boundaries
- C. Flow control
- D. Congestion control

Answer: B

5. What does IP's best-effort service mean?

- A. Guaranteed delivery
- B. Guaranteed ordering
- C. No guarantees of delivery, order, or integrity
- D. Guaranteed congestion control

Answer: C

6. What is transport-layer demultiplexing?

- A. Sending data from many sockets to the network layer
- B. Delivering received segment data to the correct socket
- C. Assigning IP addresses
- D. Encrypting application data

Answer: B

7. What is transport-layer multiplexing?

- A. Taking data from sockets and creating transport-layer segments
- B. Delivering frames to physical media
- C. Looking up DNS records
- D. Dropping packets at routers

Answer: A

8. Multiple UDP clients sending to the same destination port are directed to the same socket.

- A. True
- B. False

Answer: A

9. Multiple TCP clients sending to the same destination port are always directed to the same socket.

- A. True
- B. False

Answer: B

10. Which field is found in a UDP header?

- A. Source port number
- B. Sequence number
- C. Receiver advertised window
- D. Header length field

Answer: A

11. Why is the UDP length field needed?

- A. Because the payload can be variable length
- B. Because UDP is connection-oriented
- C. Because TCP requires it
- D. Because DNS uses it

Answer: A

12. What is a checksum used for?

- A. Detecting bit errors
- B. Assigning IP addresses
- C. Routing packets globally
- D. Opening TCP connections

Answer: A

13. A single flipped bit in one of two numbers will always change the Internet checksum.

- A. True
- B. False

Answer: A

14. A single flipped bit in each of two numbers will always change the Internet checksum.

- A. True
- B. False

Answer: B

15. In reliable data transfer, what does an ACK indicate?

- A. A packet was received correctly
- B. A packet was corrupted
- C. A packet was dropped by a router
- D. A packet has no checksum

Answer: A

16. In reliable data transfer, what does a NAK indicate?

- A. A packet was received correctly
- B. A packet was not received correctly
- C. A packet was routed successfully
- D. A TCP connection was closed

Answer: B

17. What mechanism allows duplicate detection at the receiver?

- A. Sequence numbers
- B. DNS records
- C. Cookies
- D. MAC addresses

Answer: A

18. What is a cumulative ACK(n)?

- A. It acknowledges only packet n
- B. It acknowledges all packets up to and including n
- C. It rejects packet n
- D. It closes the connection

Answer: B

19. What is the maximum utilization of stop-and-wait when packet size is 10K bits, rate is 10 Mbps, and RTT is 10 ms?

- A. 0.1
- B. 0.5
- C. 1.0
- D. 10

Answer: A

20. What is the utilization of a pipelined protocol with arbitrarily high pipelining in the same scenario?

- A. 0.1
- B. 0.5
- C. 1.0
- D. 10

Answer: C

21. How many packets can the sender transmit before receiving acknowledgments in that scenario?

- A. 1
- B. 5
- C. 10
- D. 100

Answer: C

22. Which statement about pipelining is true?

- A. Only one packet can be in flight
- B. Multiple unACKed packets can be in flight
- C. It disables ACKs
- D. It only works with UDP

Answer: B

23. Why might a GBN receiver discard out-of-sequence packets?

- A. The sender will resend them anyway
- B. They are always corrupted
- C. TCP requires it
- D. They contain no sequence number

Answer: A

24. Why does an SR receiver ACK packets with sequence numbers less than its current window?

- A. Because the sender may not have received the ACK yet
- B. Because the packet is always lost
- C. Because ACKs are not cumulative
- D. Because UDP requires it

Answer: A

25. TCP preserves application-defined message boundaries.

- A. True
- B. False

Answer: B

26. In TCP, why is an ACK often one larger than the received sequence number?

- A. The segment carried one byte, so the next expected byte is sequence number + 1
- B. TCP skips one byte randomly
- C. ACKs are always doubled
- D. The checksum adds one

Answer: A

27. TCP's EWMA EstimatedRTT depends on earlier samples.

- A. True
- B. False

Answer: A

28. What does TCP flow control use to prevent receiver overflow?

- A. Receiver advertised window
- B. DNS cache
- C. UDP length field
- D. HTTP cookie

Answer: A

29. What can trigger TCP Fast Retransmit?

- A. Three duplicate ACKs
- B. A DHCP request
- C. DNS caching
- D. HTTP GET

Answer: A

30. Which TCP message initiates a connection request?

- A. FIN
- B. SYN
- C. RESET
- D. FINACK

Answer: B

31. Which TCP message terminates a connection?

- A. SYN
- B. SYNACK
- C. FIN
- D. HTTP GET

Answer: C

32. Which TCP field contains the first byte index of data in the segment?

- A. Sequence number
- B. ACK bit
- C. Checksum
- D. Header length

Answer: A

33. Which TCP field contains the next expected byte?

- A. Source port
- B. ACK number field
- C. Data field
- D. Header checksum

Answer: B

34. If TCP sends segments with sequence numbers 100, 200, 300, 400, and 500, and segment 300 is lost, what ACK is sent after receiving segment 500?

- A. ACK 200
- B. ACK 300
- C. ACK 400
- D. ACK 600

Answer: B

35. Which action is used in network-assisted congestion control?

- A. Router marks a field in the datagram header
- B. Sender only waits for timeout
- C. Receiver disables ACKs
- D. DNS changes the IP address

Answer: A

36. Which action is associated with end-to-end congestion control?

- A. Sender reduces rate after detecting loss via ACKs
- B. Router directly controls sender window
- C. DNS sends congestion messages
- D. Web cache drops ACKs

Answer: A

37. What does AIMD do after triple duplicate ACKs?

- A. Doubles cwnd
- B. Cuts cwnd in half
- C. Sets cwnd to infinity
- D. Disables TCP

Answer: B

38. What does AIMD do after a timeout?

- A. Cuts cwnd to 1
- B. Doubles cwnd
- C. Keeps cwnd unchanged
- D. Stops using ACKs

Answer: A

39. In the absence of loss, TCP slow start increases the sending rate:

- A. Linearly
- B. Exponentially per RTT
- C. Not at all
- D. More slowly than AIMD

Answer: B

40. What is an advantage of QUIC streams?

- A. They eliminate IP
- B. One stream's error does not block other streams
- C. They require operating system updates
- D. They disable congestion control

Answer: B

Chapter 4: Network Layer — MCQs

1. Where is the network layer implemented?

- A. Hosts only
- B. Routers only
- C. Both hosts and routers
- D. DNS servers only

Answer: C

2. Which action is part of the network-layer data plane?

- A. Moving a datagram from input port to output port
- B. Computing global routing paths only
- C. Exchanging routing messages only
- D. Updating an application protocol

Answer: A

3. Which action is primarily a control-plane action?

- A. Dropping a packet from a full buffer
- B. Looking up address bits in a forwarding table
- C. Routers exchanging link-cost information
- D. Moving a datagram through a router

Answer: C

4. What quality-of-service guarantees are included in the Internet's best-effort service model?

- A. Guaranteed bandwidth
- B. Guaranteed delivery
- C. Guaranteed delay
- D. None of these guarantees

Answer: D

5. Which router component operates primarily in the control plane?

- A. Input ports
- B. Switching fabric
- C. Output ports
- D. Routing processor

Answer: D

6. Where is the destination IP address looked up in a router?

- A. At the input port
- B. At the application layer
- C. At the DNS server
- D. At the web cache

Answer: A

7. What happens if a datagram arrives at an output port with no free buffers?

- A. It is always sent back to the input port
- B. It may be dropped or another packet may be removed
- C. It is converted to TCP
- D. It is sent to DNS

Answer: B

8. What is Head-of-the-Line blocking?

- A. A front queued datagram prevents others behind it from moving forward
- B. A TCP header is too long
- C. A DNS request is cached
- D. A packet is encrypted

Answer: A

9. Which is a principal component of IPv4?

- A. IPv4 datagram format
- B. HTTP cookie format
- C. SMTP command format
- D. IMAP mailbox format

Answer: A

10. What does the IPv4 Time-to-Live field do?

- A. Increases at each router
- B. Is decremented at each router and prevents endless loops
- C. Stores the destination port number
- D. Stores the TCP ACK number

Answer: B

11. What does the IPv4 upper-layer field contain?

- A. The protocol number for TCP, UDP, etc.
- B. The DNS hostname
- C. The MAC address
- D. The HTTP response code

Answer: A

12. What is an IP address associated with?

- A. An interface
- B. A person
- C. Only a router name
- D. A web page

Answer: A

13. If a host has more than one interface, it may have:

- A. No IP address
- B. More than one IP address
- C. Only one IP address always
- D. Only a MAC address

Answer: B

14. What is an IP subnet?

- A. A group of interfaces that can reach each other without an intervening router
- B. A TCP connection
- C. A DNS cache
- D. An HTTP session

Answer: A

15. What is the maximum number of interfaces in a /24 network?

- A. 8
- B. 16
- C. 128
- D. 256

Answer: D

16. What is the maximum number of interfaces in a /29 network?

- A. 4
- B. 8

- C. 16
- D. 256

Answer: B

17. What does it mean that DHCP is “plug and play”?

- A. No manual host configuration is needed
- B. It requires static IP configuration
- C. It replaces TCP
- D. It disables DNS

Answer: A

18. What is the purpose of DHCP?

- A. To obtain an IP address for a host joining a network
- B. To send email
- C. To create HTTP cookies
- D. To perform congestion control

Answer: A

19. Which field appears in IPv6 but not IPv4?

- A. Flow label
- B. Time-to-live
- C. Header checksum
- D. Datagram length

Answer: A

20. What is destination-based forwarding?

- A. Matching on destination IP address and forwarding to the associated output port
- B. Matching only on source port
- C. Sending every packet to the controller
- D. Dropping all datagrams

Answer: A

21. What is generalized forwarding?

- A. Forwarding based only on destination IP
- B. Performing actions based on one or more packet header fields
- C. Sending only TCP packets
- D. Using only DNS records

Answer: B

22. Which field can OpenFlow 1.0 match on?

- A. IP source address
- B. HTTP page title
- C. Email subject
- D. Video chunk name

Answer: A

23. In OpenFlow, what action can be taken after matching a packet header field?

- A. Drop the packet
- B. Change the textbook
- C. Remove the Internet
- D. Disable all routers

Answer: A

24. Which device can be considered a middlebox?

- A. HTTP cache
- B. Keyboard
- C. Monitor
- D. Application message

Answer: A

25. Which protocol forms the “thin waist” of the Internet protocol stack?

- A. HTTP
- B. SMTP
- C. IP
- D. Ethernet

Answer: C

26. What does the end-to-end principle suggest?

- A. Place functionality at the network edge when it cannot be fully implemented in the network
- B. Place all functionality only inside routers
- C. Remove edge hosts
- D. Use only circuit switching

Answer: A

27. What does the Internet’s “hourglass” architecture mean?

- A. IP forms the thin waist in the middle of the protocol stack
- B. Every layer has only one protocol
- C. The application layer is the only layer
- D. TCP replaces IP

Answer: A

28. Which service has historically been regulated by the FCC in the US?

- A. Telecommunication services
- B. HTTP caching
- C. TCP sockets
- D. DNS root servers

Answer: A